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Worlds Beyond Earth: The Gas Giants

Jupiter and Saturn, the Kings of Planets

Aira Astralis

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Introduction

Of the eight planets that circle our star, two are imposters of a kind. Jupiter and Saturn wear the name of planets, and they are planets by every definition, yet they are not made of anything a lander could land on. They are spheres of hydrogen and helium, the two lightest and simplest elements, compressed into forms that exist nowhere on Earth, wrapped in clouds of ammonia, carrying storms wider than the Earth itself and systems of moons that dwarf the inner solar system's inventory. They are the dominant masses of the planetary system after the Sun, and everything else, our world included, is in their debt or under their shadow.

The two of them together hold better than nine-tenths of all the planetary mass orbiting the Sun. Jupiter alone outweighs every other planet combined, twice over. Their gravity sculpted the architecture of the young solar system, flinging small worlds, herding asteroids, and quite possibly shielding the inner planets from bombardment, or, in the more complicated reading, delivering the water of Earth's oceans in the first place. Their moons include the largest in the solar system and two of the most promising habitats for life anywhere: an ocean world with sulfur volcanoes, and a world with methane rain and a buried sea.

This book is a survey of the two great gas giants: what they are, how they formed, what their interiors, atmospheres, magnetospheres, rings, and moons reveal, how we have explored them, and what remains unknown. It is written for the reader who wants the whole picture, in plain language, with the numbers that make the picture real and the honest uncertainties left in place.

The distinction between the two kinds of giant comes first, because it organizes the whole series of worlds beyond Earth. Jupiter and Saturn are the hydrogen-helium giants: nine-tenths of their mass is the two primordial gases, and everything else is a trace. Uranus and Neptune, the ice giants of a companion volume, are built instead from the heavy volatile ices, water, ammonia, and methane, under a much thinner hydrogen blanket, and they are smaller, denser, colder, and stranger in their fields. The two classes formed from the same disks, and the difference between them is mostly a difference of timing and place: the hydrogen giants had to grow enormous before the gas went away, and only the very early, very rich, very cold zones of a disk can manage it.

The gas giants reward this kind of attention because they are not merely large; they are extreme in almost every measurable direction. The pressure at Jupiter's center is tens of millions of times the pressure at Earth's surface. Its magnetism is the biggest structure any planet carries, a bubble whose outline, if human eyes could catch it, would span more sky than the full Moon. Saturn's rings span most of the distance from Earth to the Moon and are, in places, thinner than the height of a house. Its moon Titan is the only satellite with a dense atmosphere, and its cousin Enceladus jets its ocean into space for any passing spacecraft to taste.

The story of how we learned these things is itself worth telling, because the gas giants have been the proving grounds of planetary science. Every generation of spacecraft has found the two worlds stranger than the generation before imagined: first that they were magnetically violent and radiatively hot, then that their moons were worlds, then that their interiors were nothing like the textbooks, and most recently that the textbook picture of their formation, settled for decades, must be revised.

What follows is organized as a journey from the outside in and then back out. We begin with the discovery and basic nature of the giants, follow their formation in the solar system's first millions of years, descend through their clouds into their exotic interiors, survey their fields and forces, their rings, and their great satellite courts, and end with the missions in flight and the questions still open. The gas giants have been watched by human eyes for all of recorded history, and by robot eyes for half a century, and they have not run out of surprises. Neither, we suspect, will this book's subject ever be finished.

One framing fact comes before the chapters begin, because it fixes the book's scale of stakes. The two gas giants hold, between them, more than nine-tenths of all the mass that orbits the Sun, and the smaller of them outweighs the entire rest of the system outside Jupiter combined; the Earth, in the ledger of this book, is a rounding error. The reader is therefore opening a book about the solar system's center of gravity, in the literal and financial sense of the phrase, and the chapters that follow explain how the account came to be dominated by two worlds the reader has never seen as more than points of light.

Chapter 01: Seeing the Two Wanderers

Jupiter and Saturn are among the five bright wanderers that ancient skywatchers tracked without instruments, and

they were known to every literate civilization. Jupiter is, after the Sun, the Moon, and Venus, the brightest thing in the sky that does not flicker, and Saturn, dimmer and slower, held the outermost of the visible planets for every culture that kept a calendar. The names we use are Roman, but the attention was universal: the Babylonians logged Jupiter's positions for centuries, and Chinese court astronomers recorded the conjunctions and occultations of both worlds as matters of state.

The squabble, told properly, belongs to Jupiter's moons. A cleric and astronomer working in northern Italy published his observations of the four Medicean stars in 1610, and the claim that a planet had attendants struck at the Earth-centered cosmos more sharply than any orbit calculation; a colleague in Prague, Simon Marius, claimed to have seen them first, and the priority dispute, settled in the Italian's favor by notebooks and publication dates, gave the moons the names they carry today, from the Marius side, after the lovers of Zeus. The deeper shock, though, is what the discovery did to the word planet itself: a wanderer with a court of its own was a world in a fuller sense than any argument had yet claimed, and the recognition that the solar system was a system, hierarchical and dynamic, dates from those first weeks of 1610.

The telescopic era began with a squabble and a shock. Galileo, turning his instrument on Jupiter in 1610, found four small stars in a line beside the planet, and within weeks of watching them, understood they moved around it: the first known moons of another world, and the first demonstration that not everything circled Earth. He also, peering at Saturn, saw its famous handles, the edges of the rings, without understanding them; his telescope resolved the planet as a triple body that later observers watched open and close and even vanish, a cycle that took decades to explain. The explanation, when Christiaan Huygens supplied it in 1656, was a flat ring, tilted to our view, orbiting in step with the planet's seasons.

The centuries that followed were a patient accumulation. Telescopes grew, and Jupiter's cloud bands resolved into structure, its Great Red Spot was identified as a persistent feature in the 1830s, and its rapid rotation was measured from the swing of its markings. Saturn's rings were found to be multiple, then to be made of innumerable separate particles, a conclusion forced by the mathematics of their stability. The moons accumulated: Saturn's Titan, recognized by Huygens as remarkable, then Iapetus, Rhea, Dione, Tethys, then the crowded field of smaller satellites; Jupiter's four joined by an outer family discovered into the twentieth century. By the opening of the space age, the two giants were the best-studied planets after Mars and Venus, and yet almost nothing was known about them as worlds. Their visible faces were weather, and their interiors were arithmetic.

Spectroscopy, the analysis of their light, gave the first physical portrait, and it was a shock. The giants showed the signatures of hydrogen and methane and ammonia, the compounds of the lightest elements, with no hint of a solid surface, no oxygen-rich chemistry, nothing to land on. Their densities, computable from their sizes and masses, were barely more than water's. The conclusion, worked out in the early twentieth century and never seriously revised, is that Jupiter and Saturn are not rocky planets with thick blankets but a different kind of body altogether: mostly hydrogen and helium throughout, compressing under their own weight into liquid metallic states in their deep interiors, with the rock and ice of a terrestrial world's worth of material sunk invisibly into the middle. They are, in the phrase this book will use carefully, failed stars only in the loosest sense: bodies of a star's composition, lacking a star's mass, and never having come close to igniting.

The discovery chapter of the gas giants thus closed in the 1950s with the essentials known and almost everything else open. The sizes, masses, densities, compositions, rotations, and moons were established by generations of patient ground-based work. What the giants were like inside, how they were built, what forces ran through them, and whether their moons might matter to the question of life were all unknown, and each of those questions would need spacecraft to answer. The era of robot explorers opened with the giants at the top of the target list, for a reason that was both scientific and practical: they were far, they were huge, and they were on the way out of the solar system for any mission with enough speed.

The two wanderers moved on schedules a patient scribe could set a calendar by, and the ancients did. Jupiter laps the zodiac in just under twelve years, crossing one constellation's worth of sky each year, the cadence that still fixes the twelve-sign logic of astrology; Saturn takes nearly thirty, drifting one sign every two and a half years, so a single human lifetime holds little more than one of its circuits. The two bright ones were thus the sky's long clocks, the fast and the slow, and their meetings, the years when the slower is overtaken, arrive once in two decades and were tracked across the whole of recorded astronomy. A modern reader can run the same experiment with binoculars and patience: mark Jupiter against the stars one winter, and the mark will have shifted by one zodiac span by the next.

The discovery chapter closes on the ancients' half of the ledger, because the giants were known as planets for as

long as the word has existed, and the pre-telescopic observers who tracked them earned their exact due: not lucky, but skilled, able to predict the wanderers' returns to the precision their naked eyes and clay tablets allowed, and the Gas Giants' motions were thus part of the human record for millennia before anyone knew there were two of them, what they were, or that the class would eventually include the exoplanet hundreds the final chapters describe. The book's subject, in other words, is one of the oldest things humanity has watched and one of the newest things it understands.